

iFood Grocery Retail Store

BY ALLEY TRAN



BUSINESS BACKGROUND & CONTEXT

About Our Company, iFood

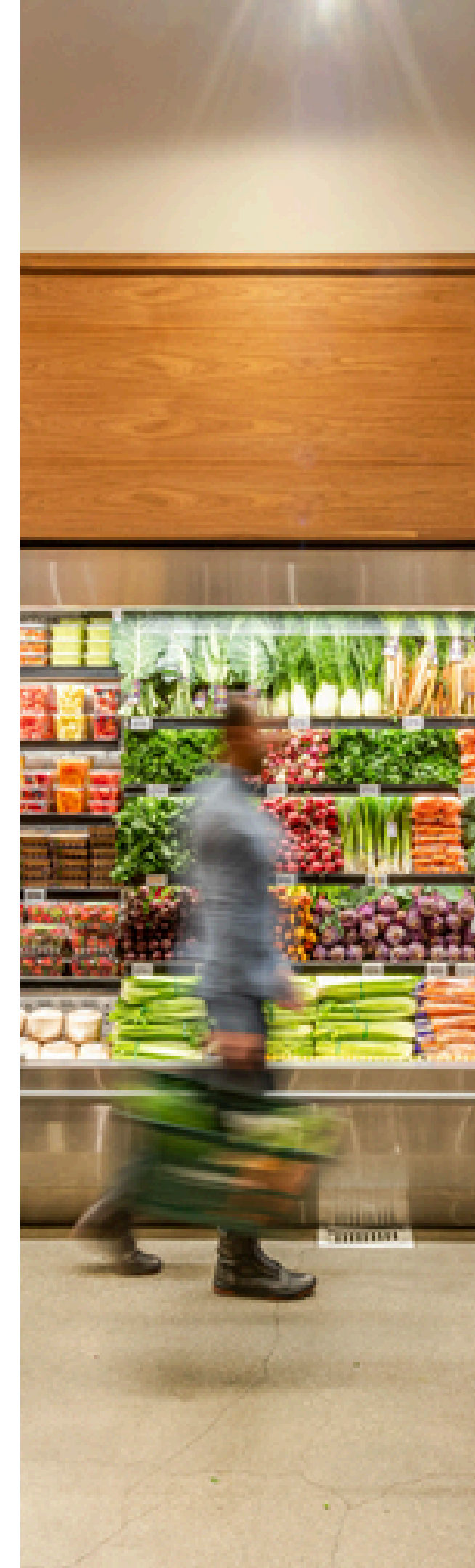
- We are a well-established retail food company
- Our portfolio spans five premium categories
 - Wines, rare meat, exotic fruits, specially prepared fish, and confectionery
- Products are acquired through three major sales channels
 - Physical stores, catalogs, and the e-commerce
- We maintain both gold-tier (premium) and regular product lines to serve diverse customer segments

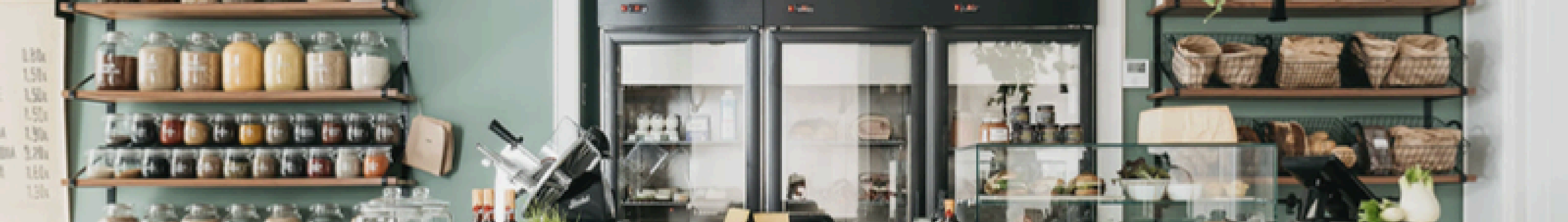
Recent Performance

- Solid revenues and a healthy bottom line in the past 3 years

Current Challenge

- Our profit outlook for the next three years is not promising





STRATEGIC INITIATIVE

- One of the strategic pillars is **improving the performance of marketing activities**, with a special focus on marketing campaigns and budget optimization.
- **Activity:** launch a new direct Marketing campaign that aims at selling a new kitchen gadget to iFood existing customers.
- **Plan:** launch a pilot campaign advertising this gadget to 2,240 customers, selected randomly. Using the result of this pilot campaign and the customer's socio-demographic information, build a predictive model that will carefully select the right customers included in the next official direct mail campaign.

PROJECT OBJECTIVE

1

Prediction and Segmentation

Build a predictive model that identifies the characteristics and behaviors of the most responsive and profitable customers to iFood marketing campaigns.

2

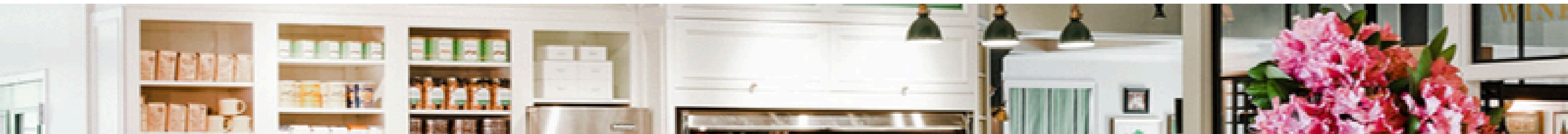
Profit Maximization

Improve the performance and maximize profits of the next direct Marketing campaign.

3

Quantitative Decision Making

Prove the effectiveness of a quantitative approach to Marketing decision making and investment.



DESCRIPTION OF THE DATABASE

The data set contains socio-demographic and firmographic features about 2,240 customers who were contacted. Additionally, it contains a flag for those customers who responded to the campaign by buying the product.

- Response (target) – 1 if customer accepted the offer in the last campaign, 0 otherwise
- ID – Unique ID of each customer
- Year_Birth – Age of the customer
- Complain – 1 if the customer complained in the last 2 years
- Dt_Customer – date of customer's enrollment with the company
- Education – customer's level of education
- Marital – customer's marital status
- Kidhome – number of small children in the customer's household
- Teenhome – number of teenagers in the customer's household



- Income – customer's yearly household income
- MntFishProducts – the amount spent on fish products in the last 2 years
- MntMeatProducts – the amount spent on meat products in the last 2 years
- MntFruits – the amount spent on fruits products in the last 2 years
- MntSweetProducts – amount spent on sweet products in the last 2 years
- MntWines – the amount spent on wine products in the last 2 years
- MntGoldProds – the amount spent on gold products in the last 2 years
- NumDealsPurchases – number of purchases made with a discount
- NumCatalogPurchases – number of purchases made using catalog (buying goods to be shipped through the mail)
- NumStorePurchases – number of purchases made directly in stores
- NumWebPurchases – number of purchases made through the company's website
- NumWebVisitsMonth – number of visits to the company's website in the last month
- Recency – number of days since the last purchase
- AcceptedCmp1 – 1 if customer accepted the offer in the 1st campaign, 0 otherwise
- AcceptedCmp2 – 1 if customer accepted the offer in the 2nd campaign, 0 otherwise
- AcceptedCmp3 – 1 if customer accepted the offer in the 3rd campaign, 0 otherwise
- AcceptedCmp4 – 1 if customer accepted the offer in the 4th campaign, 0 otherwise
- AcceptedCmp5 – 1 if customer accepted the offer in the 5th campaign, 0 otherwise





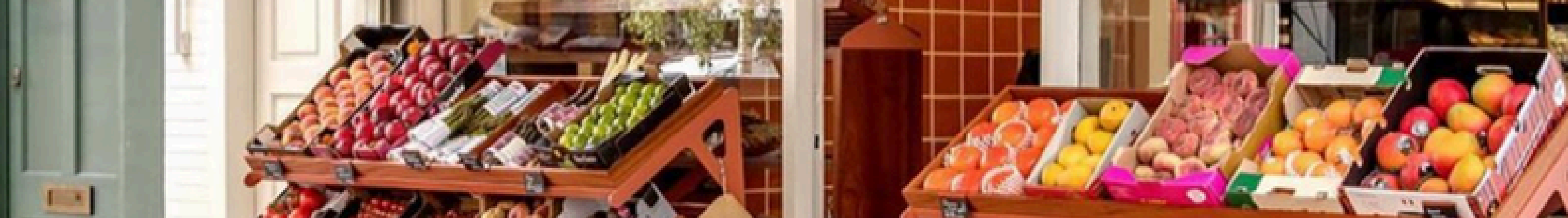
Training Sample

ID	Year_Birth	Education	Marital_Stat us	Kidhome	Income	Teenhome	Recency	Amount	Purchases	AcceptedCm p1	AcceptedCm p2	AcceptedCm p3	AcceptedCm p4	AcceptedCm p5	Total # of campaigns	Response	Complain
2174	1954	Graduation	Single	1	46344	1	38	27	6	0	0	0	0	0	0	0	0
4141	1965	Graduation	Together	0	71613	0	26	776	21	0	0	0	0	0	0	0	0
6182	1984	Graduation	Together	1	26646	0	26	53	8	0	0	0	0	0	0	0	0

Holdout Sample

ID	Year_Birth	Education	Marital_Stat us	Kidhome	Income	Teenhome	Recency	Amount	Purchases	AcceptedCm p1	AcceptedCm p2	AcceptedCm p3	AcceptedCm p4	AcceptedCm p5	Total # of campaigns	Response	Complain
5524	1957	Graduation	Single	0	58138	0	58	1617	25	0	0	0	0	0	0	1	0
5324	1981	PhD	Married	1	58293	0	94	422	19	0	0	0	0	0	0	0	0
7446	1967	Master	Together	0	62513	1	16	716	22	0	0	0	0	0	0	0	0

Amount (Monetary) = MntFishProducts + MntMeatProducts + MntFruits + MntSweetProducts + MntWines + MntGoldProds
Purchases (Frequency) = NumDealsPurchases + NumCatalogPurchases + NumStorePurchases + NumWebPurchases
Total # of Campaigns = AcceptedCmp1 + AcceptedCmp2 + AcceptedCmp3 + AcceptedCmp4 + AcceptedCmp5



PROFIT CALCULATION BASED ON CUSTOMER LIFETIME VALUE

Sample size	1120
Solicitation Cost	\$ 3.00
CLV	\$ 15.00
Cutoff response rate	20%
Scenario 1	Sending mailers to the entire Holdout sample
Number of actual sales based on 1120 customers	167
Profit from 1120 customers	\$ (855.00)

RFM Analysis

Training Dataset

RFM	Average of Response	Target (>20%)
1 1 1	0.136	FALSE
1 1 2	0.418	TRUE
1 2 1	0.273	TRUE
1 2 2	0.224	TRUE
2 1 1	0.041	FALSE
2 1 2	0.189	FALSE
2 2 1	0.093	FALSE
2 2 2	0.189	FALSE

Breakeven	Actual Sales	Profit
20%	67	\$360

Holdout Dataset

RFM	Number of People	Number of Purchases	Average Response Rate	Target (based on testing)
1 1 1	357	52	0.146	FALSE
1 1 2	71	29	0.408	TRUE
1 2 1	41	6	0.146	TRUE
1 2 2	103	32	0.311	TRUE
2 1 1	328	16	0.049	FALSE
2 1 2	82	12	0.146	FALSE
2 2 1	34	1	0.029	FALSE
2 2 2	104	19	0.183	FALSE

Limitation

The Training dataset of 1120 observations is too small to get highly accurate segmentation.
8 RFM segments carry the risk of generalization when applied to a much bigger customer database.



Modeling Process

1. Modeling Method **List Scoring**

2. Customer Lifetime Value

3. Models and Conclusion



MODELING METHOD

LIST SCORING

Goal

1. Identify the attributes (variables) that most accurately predict the probability of a customer accepting the last campaign offer (Response=1)
2. Identify the attributes and model with the highest Profit (vs targeting the entire customer database)

Process

1. Randomly divide the dataset into Training and Holdout samples with 1120 observations each.
2. In the Training sample, test several models via Binary Logistic Regression (Logit), calculate the incremental profit, and compare it to the baseline of targeting the entire group.

COEFFICIENT SUMMARY

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-10.6345	17.0298	-0.6245	0.5323
Year_Birth	0.0042	0.0086	0.4853	0.6275
EducationBasic	-0.9296	1.186	-0.7838	0.4332
EducationGraduation	0.9444	0.4452	2.1213	0.0339 *
EducationMaster	1.2305	0.486	2.5316	0.0114 *
EducationPhD	1.4919	0.4746	3.1436	0.0017 **
Income	-0.000019135	0.0000087698	-2.182	0.0291 *
Kidhome	0.352	0.2491	1.4129	0.1577
Teenhome	-0.8365	0.2397	-3.4906	0.0005 ***

	Estimate	Std. Error	z value	Pr(> z)
Recency	-0.0261	0.0039	-6.7538	1.4403e-11 ***
Amount	0.0009	0.0003	2.7477	0.0060 **
Purchases	0.0431	0.0203	2.1215	0.0339 *
AcceptedCmp3	2.1767	0.3061	7.1119	1.1445e-12 ***
AcceptedCmp4	0.8314	0.3559	2.3358	0.0195 *
AcceptedCmp5	1.3997	0.3606	3.8819	0.0001 ***
AcceptedCmp1	1.2187	0.3593	3.3918	0.0007 ***
AcceptedCmp2	1.4605	0.6272	2.3288	0.0199 *
Complain	-0.5029	1.3653	-0.3683	0.7126

LOGISTIC REGRESSION MODEL

Logistic Regression code on Test Sample

```
model <- glm(Response ~ Income + Teenhome + Recency + Amount +  
  AcceptedCmp1 + AcceptedCmp2 + AcceptedCmp3 + AcceptedCmp4 + AcceptedCmp5,  
  data = campaign, family = "binomial")
```

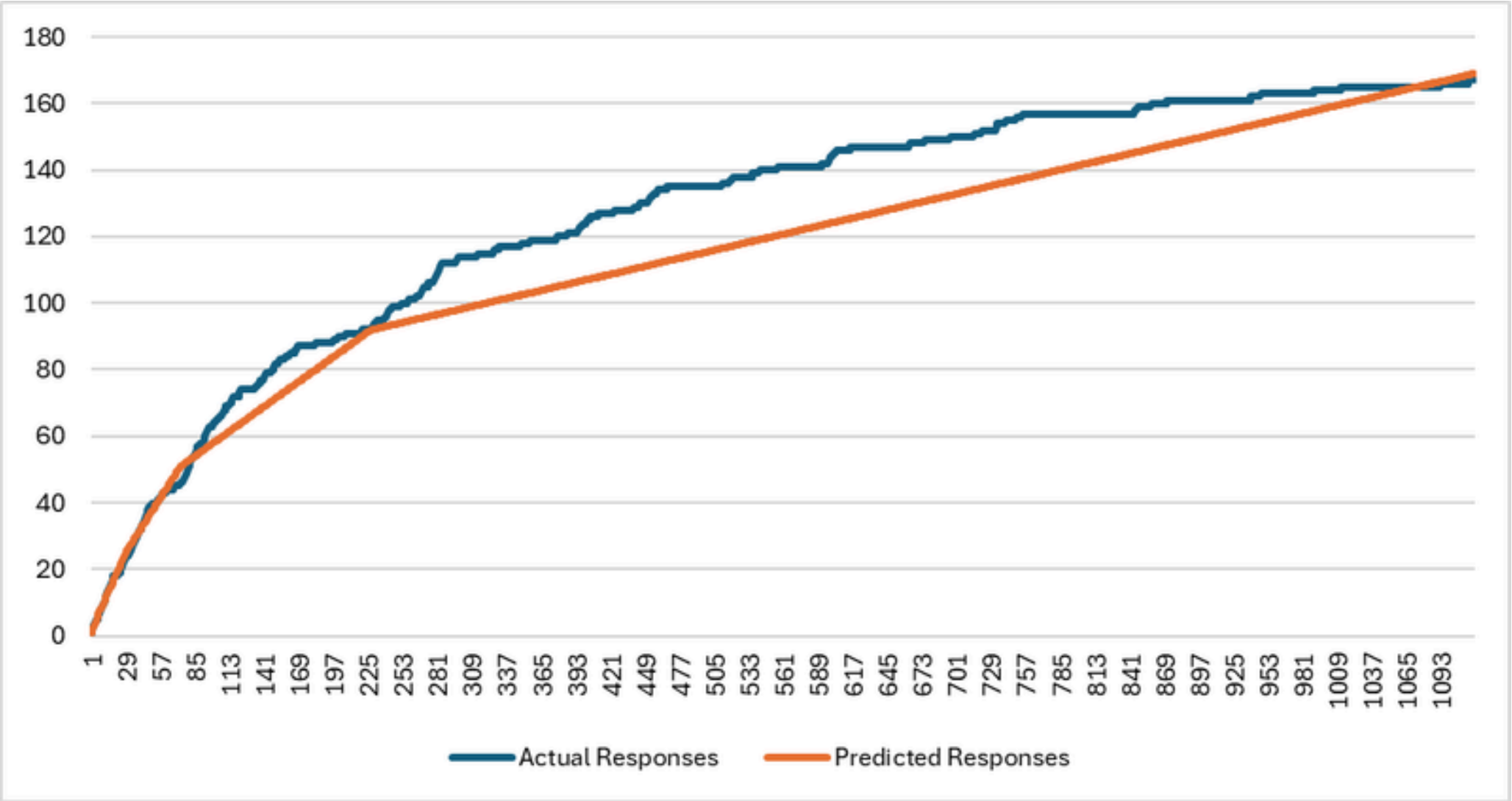
Score and Probability calculation on Holdout Sample

Income	Teenhome	Recency	Amount	Purchases	Response	AcceptedCmp1	AcceptedCmp2	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	Score	Probability	Actual Purchases	Predicted Purchases
83512	0	31	2157	22	1	1	1	1	0	1	5	0.99	1	0.99
65169	0	23	1189	28	1	1	1	1	0	1	4.45	0.99	2	1.98
84865	0	1	1688	16	1	1	1	0	1	1	4.09	0.98	3	2.97
85683	0	6	1789	17	1	1	1	0	1	1	4.06	0.98	4	3.95
80134	0	11	1378	21	0	1	1	0	1	1	3.59	0.97	4	4.92
81205	0	43	2009	19	1	0	1	1	0	1	3.51	0.97	5	5.89
87679	0	62	2524	29	1	1	0	1	0	1	3.17	0.96	6	6.85
102692	0	5	1112	29	1	1	1	0	1	1	3.14	0.96	7	7.81
71670	0	8	1766	15	1	0	1	0	1	1	3.11	0.96	8	8.77

Model 1

Response ~ Total # of campaigns

	Estimate
(Intercept)	-2.36
Total # of campaigns	1.35
SSE	116.64

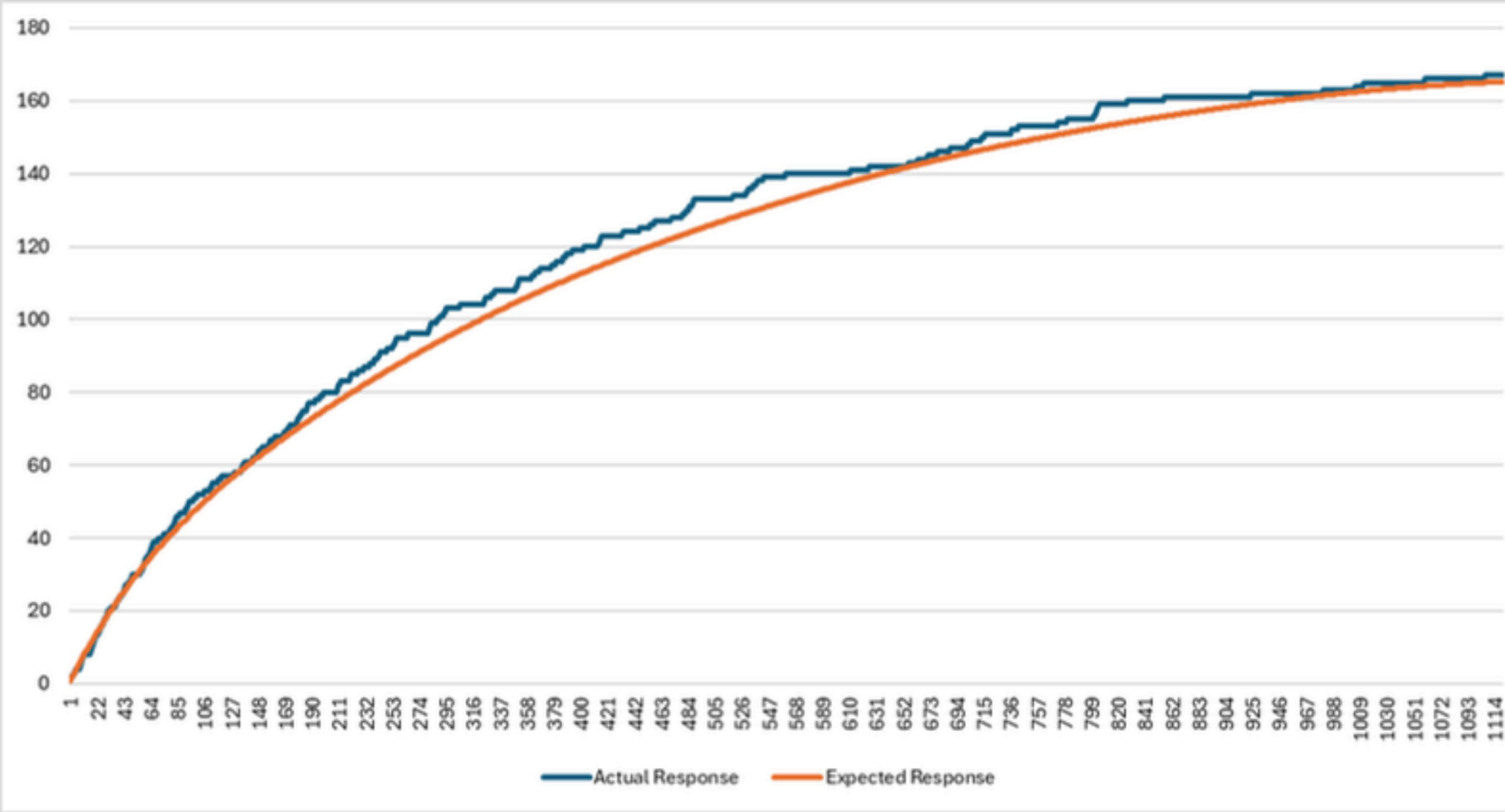


Scenario 2: Sending the mailers to only qualifying customers	
Number of customers to be targeted	227
% of customers targeted in holdout list	20.27%
# of actual sales from targeted customers	92
Profit from targeted customers	\$ 699.00
The difference in Profit between List Scoring targeting vs entire sample	\$ 1,554.00

Model 2

$$\text{Response} \sim \text{Income} + \text{Teenhome} + \text{Recency} + \text{Amount}$$

	Estimate
(Intercept)	-0.929
Teenhome	-0.746
Recency	-0.0206
Income	-0.0000113
Amount	0.00136
SSE	120.57



Scenario 2: Sending the mailers to only qualifying customers

Number of customers to be targeted270

% of customers targeted in holdout list24.11%

of actual sales from targeted customers96

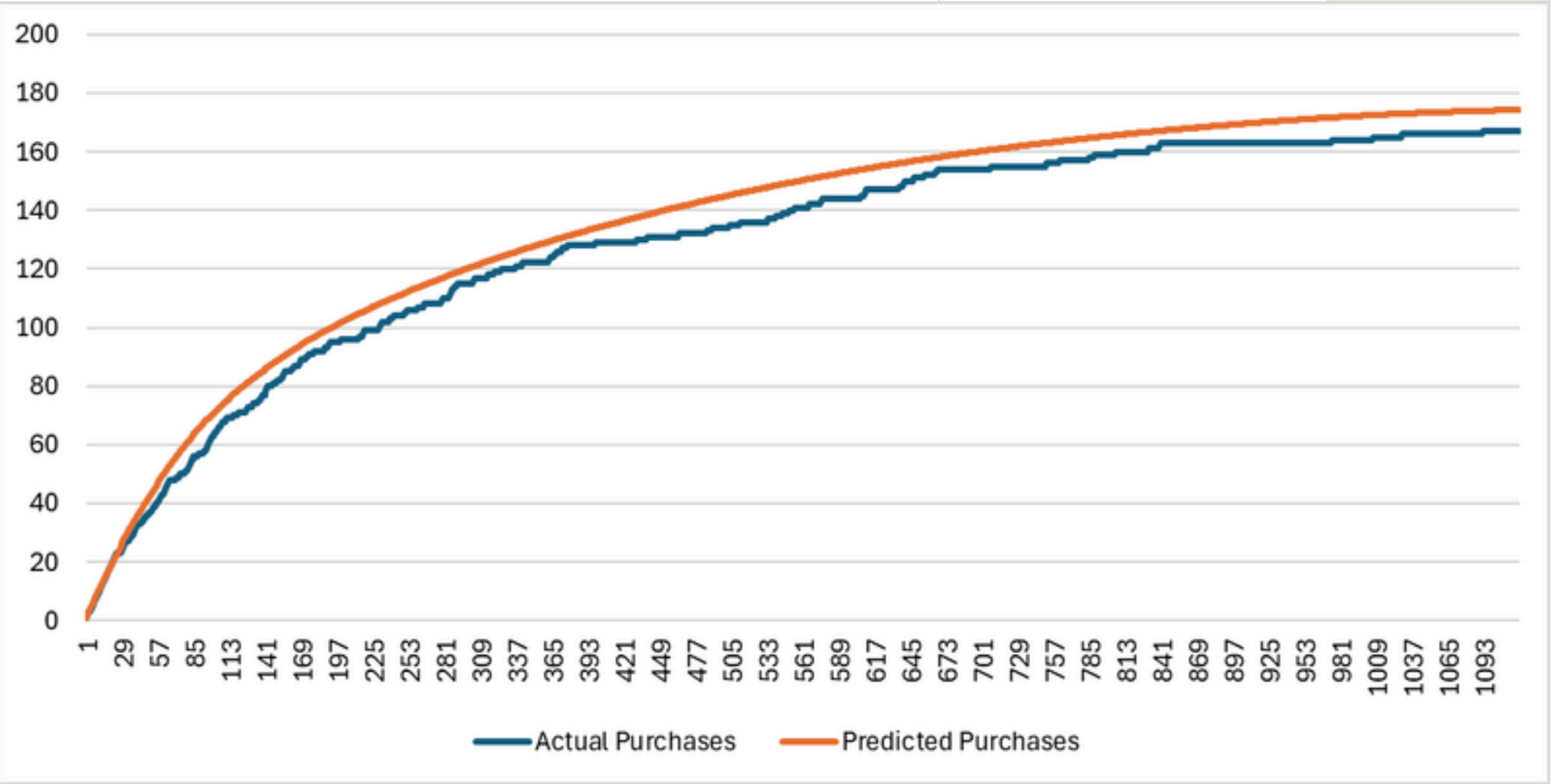
Profit from targeted customers\$630.00

The difference in Profit between List Scoring targeting vs entire sample\$1,485.00

Model 3

**Response ~ Income +
Teenhome + Recency +
Amount +
Total # of campaigns**

	Estimate
(Intercept)	-0.695
Income	-0.0000184
Teenhome	-0.672
Recency	-0.0248
Amount	0.000986
Total # of campaigns	1.29
SSE	107.22



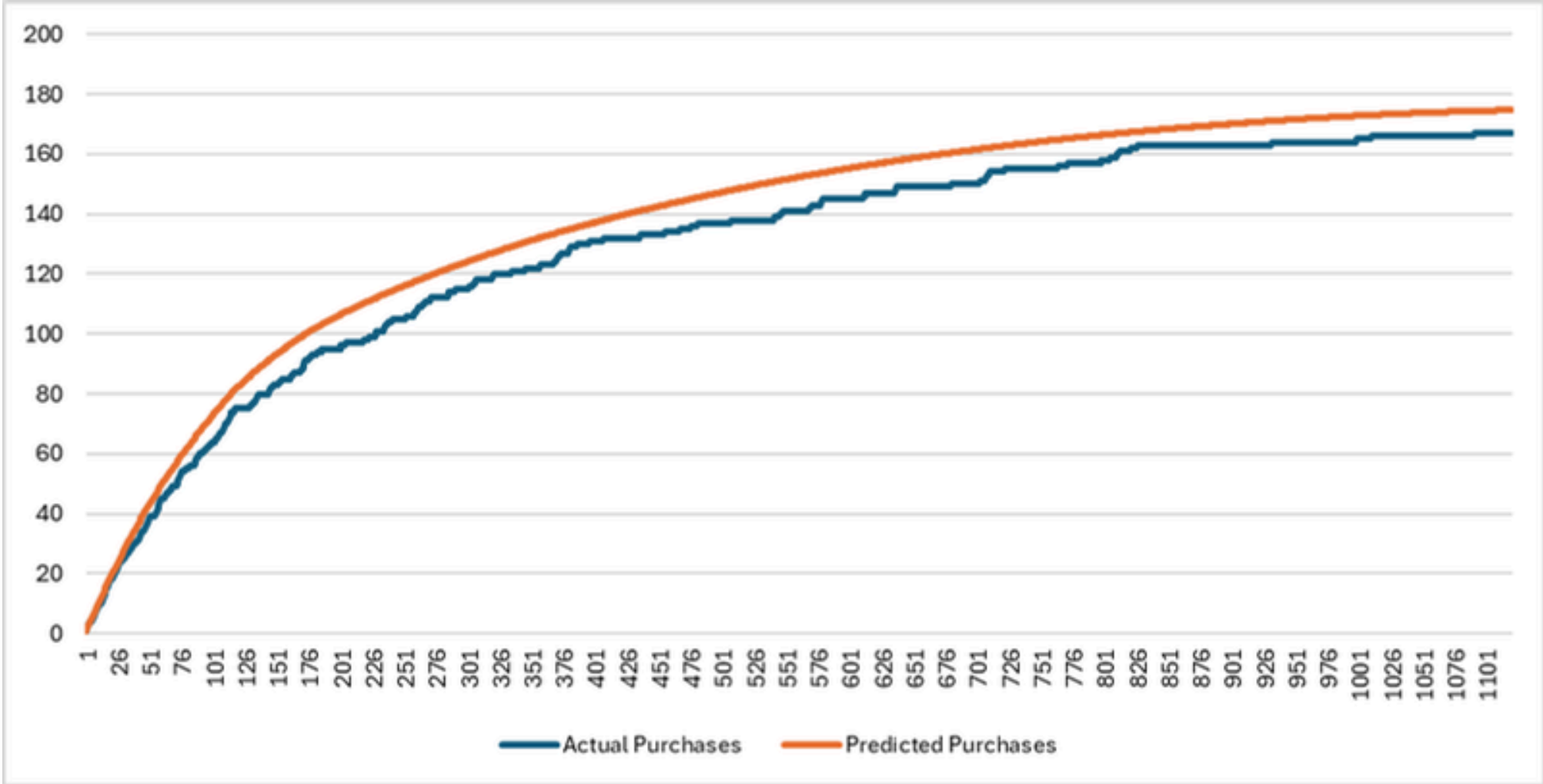
**Scenario 2: Sending the mailers to only
qualifying customers**

Number of customers to be targeted	227
% of customers targeted in holdout list	20.27%
# of actual sales from targeted customers	99
Profit from targeted customers	\$ 804.00
The difference in Profit between List Scoring targeting vs entire sample	\$ 1,659.00

Model 4

Response ~ Income + Teenhome + Recency + Amount + Cmp1 + Cmp2 + Cmp3 + Cmp4 + Cmp5

	Estimate
(Intercept)	-1.01
Income	-0.0000143
Teenhome	-0.643
Recency	-0.0246
Amount	0.00104
AcceptedCmp1	1.08
AcceptedCmp2	1.39
AcceptedCmp3	2.05
AcceptedCmp4	0.91
AcceptedCmp5	1.21
SSE	105.82



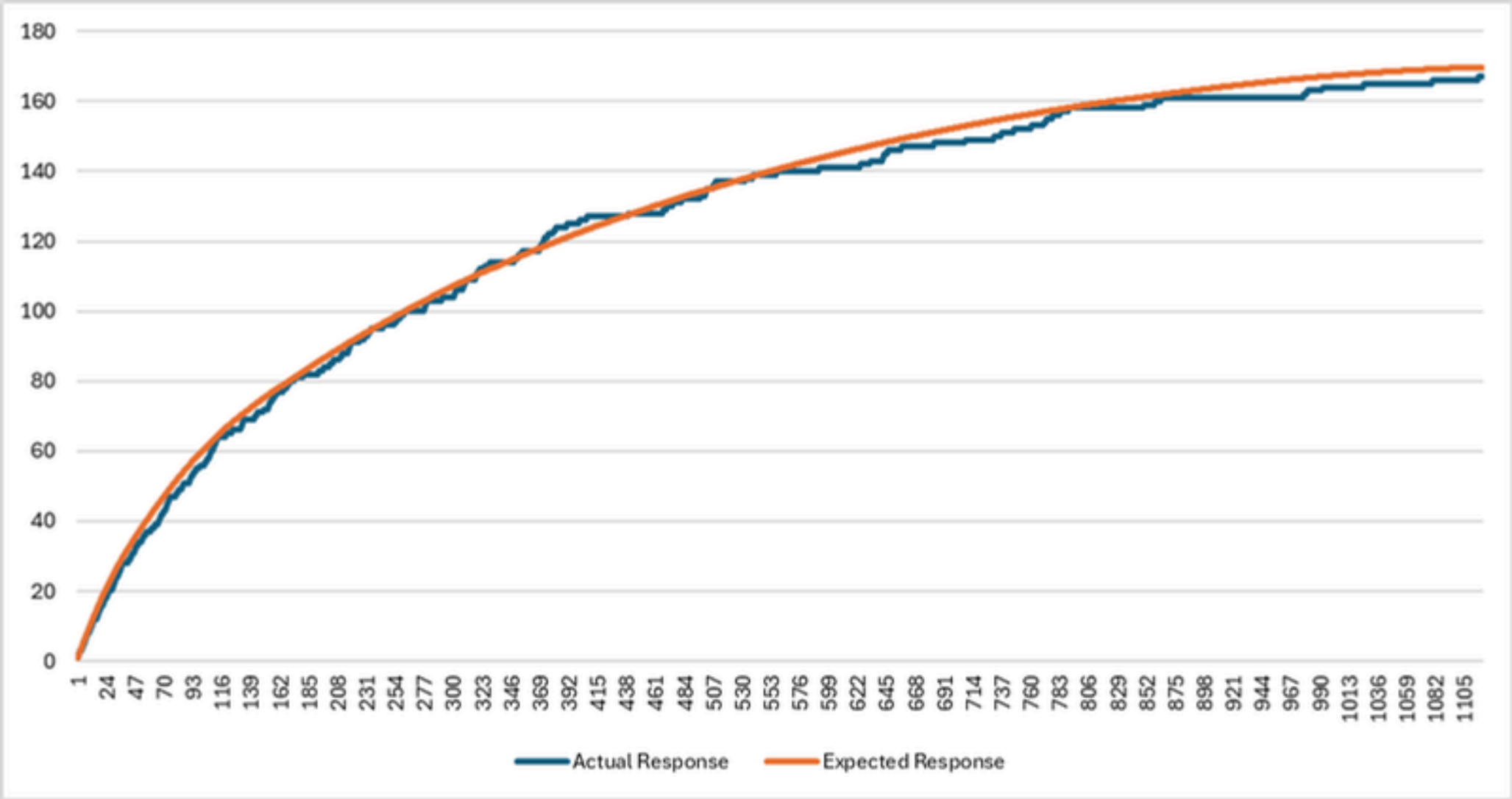
Scenario 2: Sending the mailers to only qualifying customers	
Number of customers to be targeted	214
% of customers targeted in holdout list	19.11%
# of actual sales from targeted customers	97
Profit from targeted customers	\$ 813.00
The difference in Profit between List Scoring targeting vs entire sample	\$ 1,668.00

Model 5

**Response ~ Income +
Teenhome + Recency +
Amount + Cmp1 +
Cmp2**

	Estimate
(Intercept)	-0.672
Teenhome	-0.722
Recency	-0.0225
Income	-0.0000172
Amount	0.00127
AcceptedCmp1	1.53
AcceptedCmp2	2.37
SSE	114.74

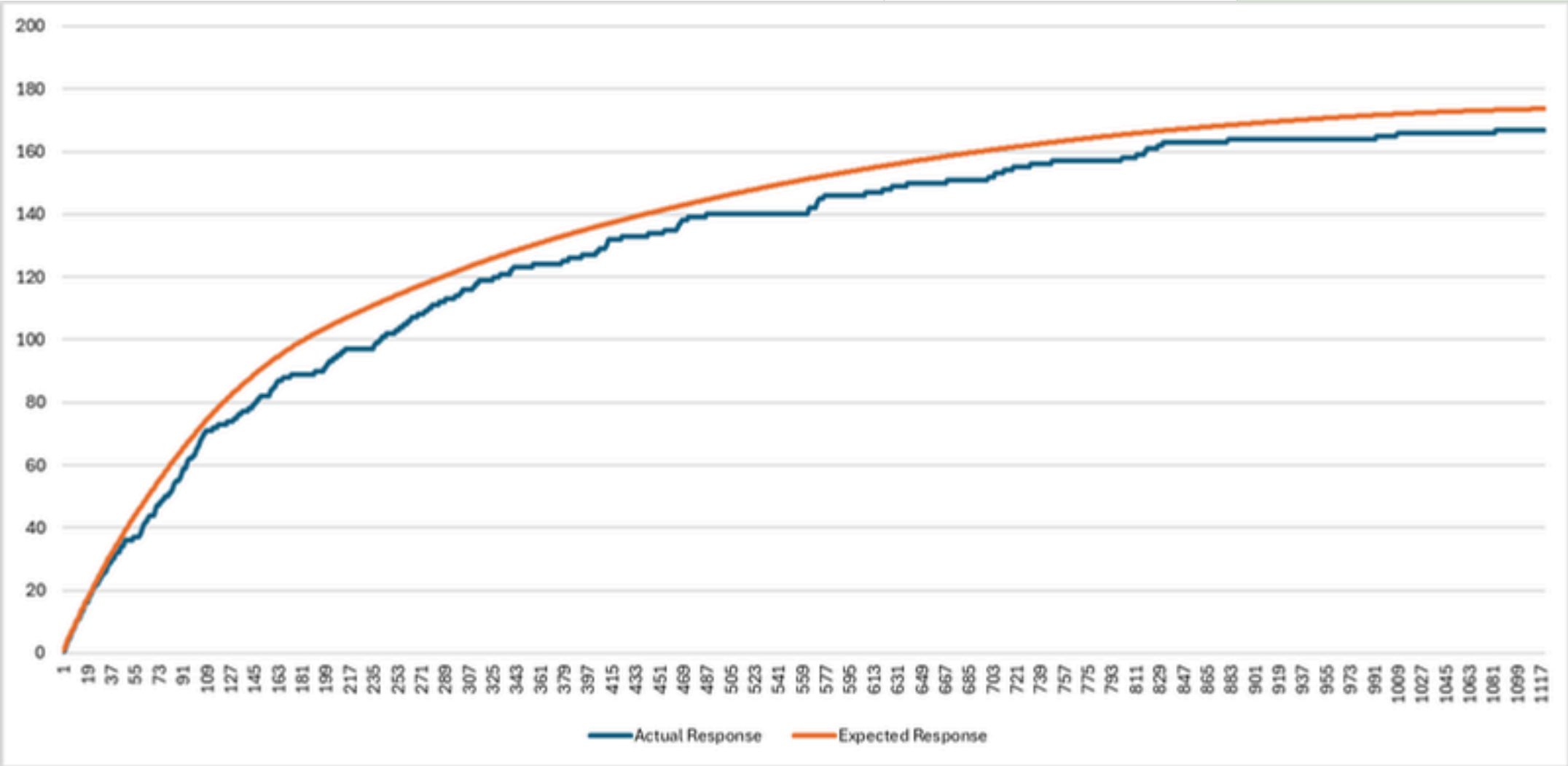
Scenario 2: Sending the mailers to only qualifying customers	
Number of customers to be targeted	241
% of customers targeted in holdout list	21.52%
# of actual sales from targeted customers	95
Profit from targeted customers	\$702.00
The difference in Profit between List Scoring targeting vs entire sample	\$1,557.00



Model 6

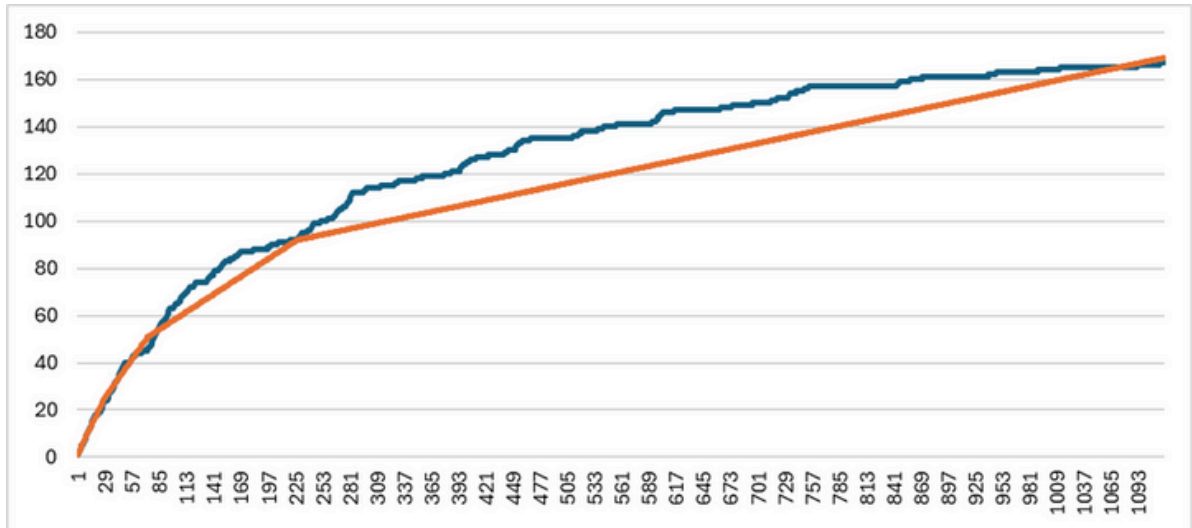
$$\text{Response} \sim \text{Income} + \text{Teenhome} + \text{Recency} + \text{Amount} + \text{Cmp3} + \text{Cmp4} + \text{Cmp5}$$

	Estimate
(Intercept)	-1.14
Teenhome	-0.697
Recency	-0.0245
Income	-0.0000104
Amount	0.00103
AcceptedCmp3	2.17
AcceptedCmp4	1.3
AcceptedCmp5	1.32
SSE	107.67



Scenario 2: Sending the mailers to only qualifying customers	
Number of customers to be targeted	217
% of customers targeted in holdout list	19.38%
# of actual sales from targeted customers	97
Profit from targeted customers	\$804.00
The difference in Profit between List Scoring targeting vs entire sample	\$1,659.00

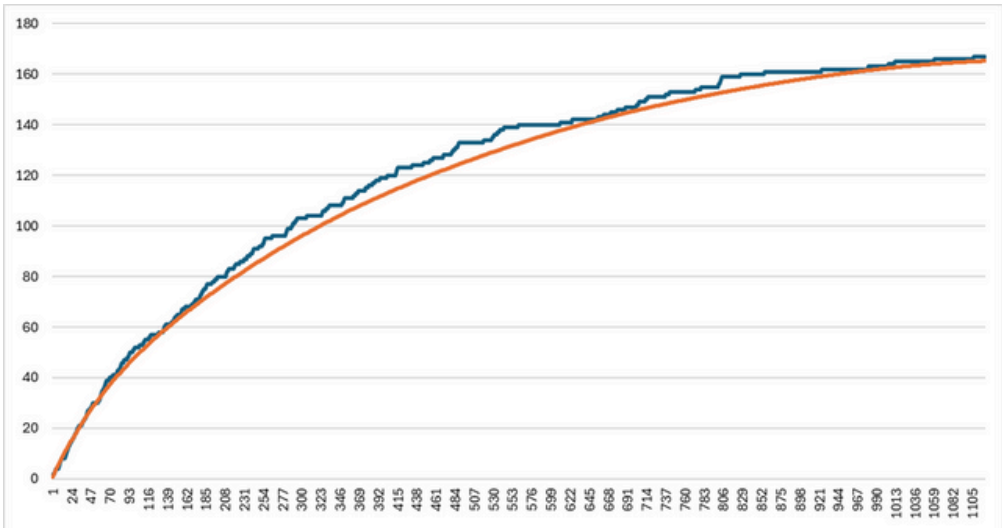
Incremental Profit per Model



Model 1

Response ~ Total # of campaigns

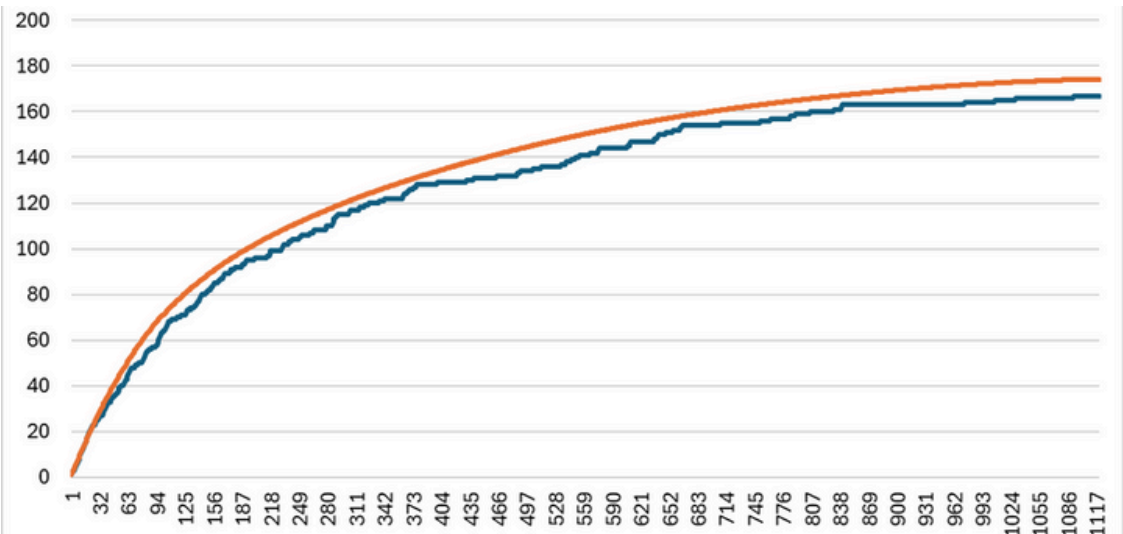
\$ 1,554



Model 2

Response ~ Income + Teenhome + Recency + Amount

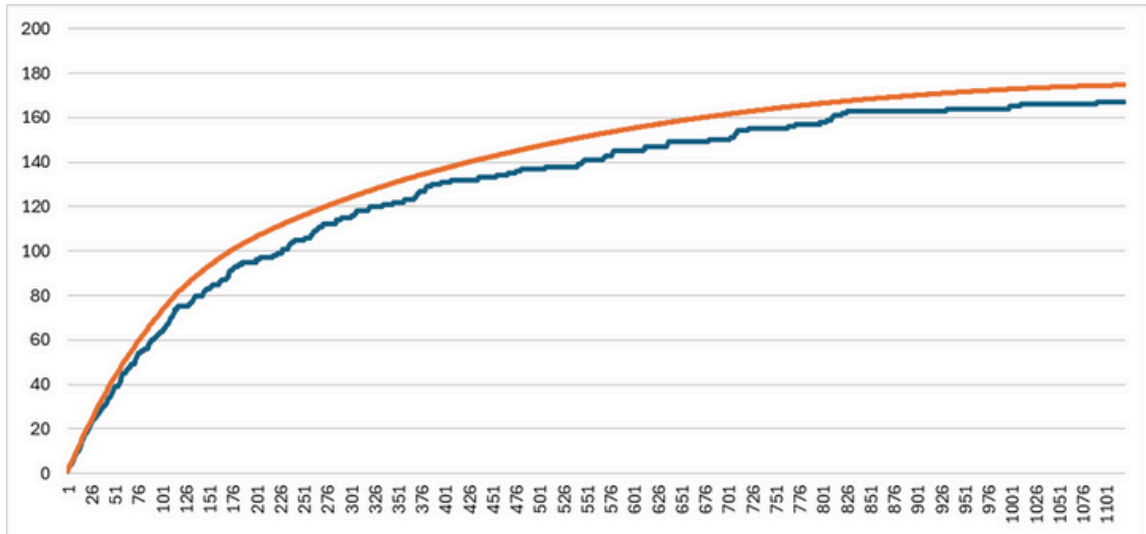
\$1,485



Model 3

Response ~ Income + Teenhome + Recency + Amount + Total # of campaigns

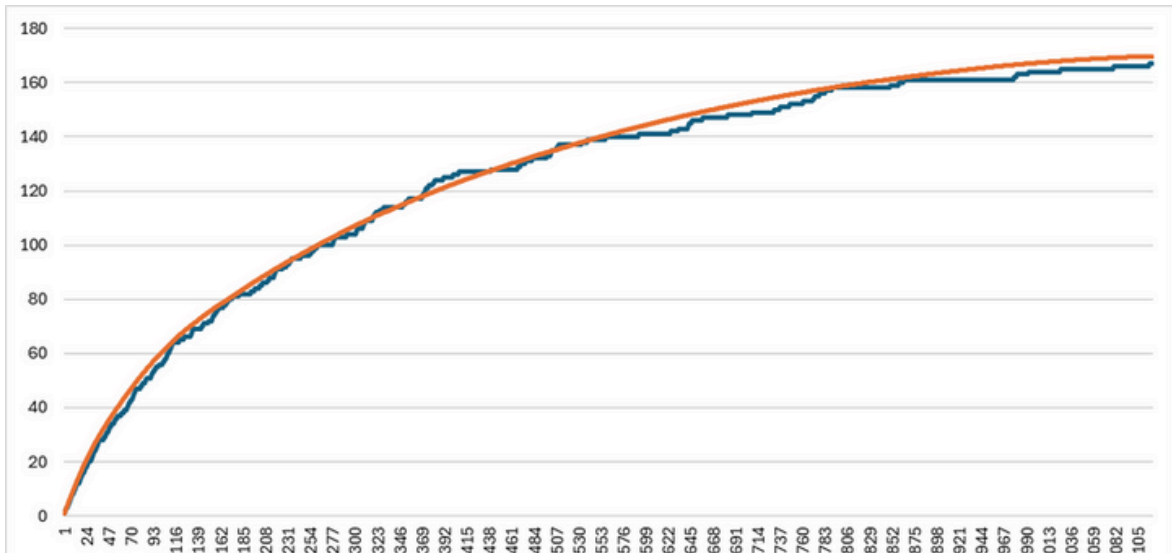
\$ 1,659



Model 4

Response ~ Income + Teenhome + Recency + Amount + Cmp1 + Cmp2 + Cmp3 + Cmp4 + Cmp5

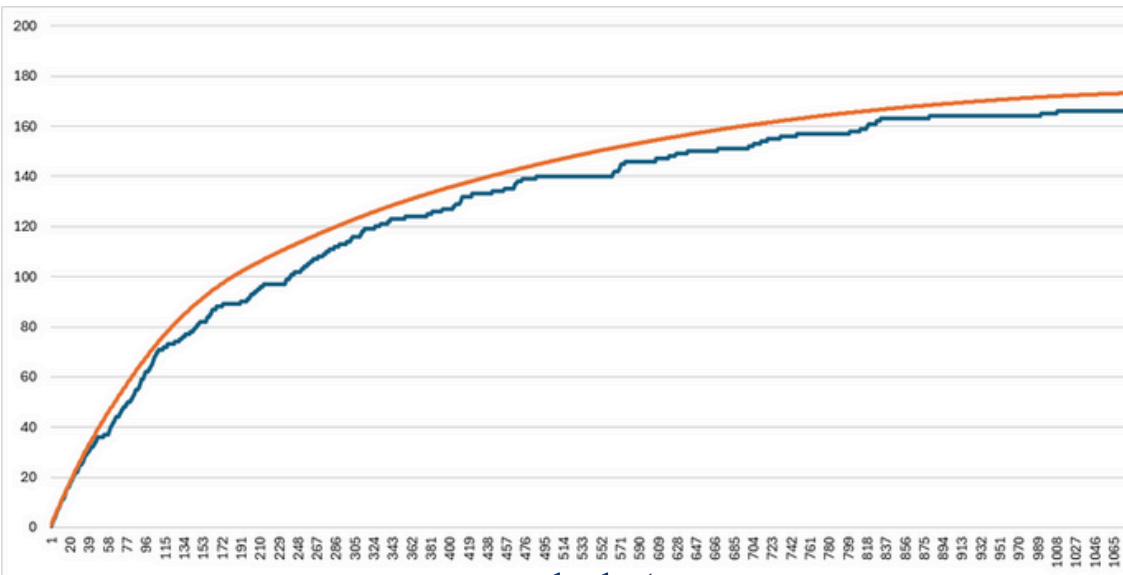
\$1,668



Model 5

Response ~ Income + Teenhome + Recency + Amount + Cmp1 + Cmp2

\$1,557



Model 6

Response ~ Income + Teenhome + Recency + Amount + Cmp3 + Cmp4 + Cmp5

\$1,659



INSIGHTS

1. Within this database, **attributes of statistical significance** in determining the probability of accepting a campaign are:
 - a. Income
 - b. Teenhome (number of teenage children)
 - c. Recency (number of days from last purchase)
 - d. Amount (total amount of money spent in the last 2 years)
 - e. AcceptedCmp1 – 5 (accepted offer in any of the campaign)
2. The **most important variable** in determining the acceptance of a campaign is **Campaign 3** result.
3. Among several models tested, the **model with the highest accuracy** in determining probability of purchases (based on SSE) are:
 - a. Income + Teenhome + Recency + Amount + Cmp1 + Cmp2 + Cmp3 + Cmp4 + Cmp5
 - b. Income + Teenhome + Recency + Amount + Total # of campaigns
4. Based on Incremental Profit between Targeted Campaign vs Total Sample, the model with the **highest incremental profit** is:
Income + Teenhome + Recency + Amount + Cmp1 + Cmp2 + Cmp3 + Cmp4 + Cmp5

RECOMMENDATIONS

1

Apply List Scoring model on the entire customer database from the last 2 years with independent variables: Income, Teenhome, Recency, Amount, Campaign Acceptance

2

Target customers with Probability of Campaign Acceptance above the Cutoff Threshold of 20%

3

Study the difference between Campaign 3 and other campaigns. Apply learnings from campaign 3 to the new campaign to increase effectiveness

